

Eg  $\Lambda = ab$  group. The constant assignment

$$\text{top space } X \ni U \xrightarrow{\text{open}} \Lambda$$

is not a sheaf in general since if  $U$  and  $V$  are disjoint opens and  $a \neq b \in \Lambda$ , then  $\exists x \in \Lambda$  with

$$\text{res}_U^{U \cup V}(x) = a \quad \text{res}_V^{U \cup V}(x) = b$$

But  $U \mapsto \Lambda^{\pi_0(U)}$ ,  $\pi_0(U) = \text{set of connected components of } U$

$$= \{f: U \rightarrow \Lambda \text{ sts}\} \text{ with } \Lambda \text{ discrete,}$$

is a sheaf called the constant sheaf assoc to  $\Lambda$ , denoted  $\underline{\Lambda}$ .

Fact The forgetful functor  $\text{Shv}(X) \rightarrow \text{PreShv}(X)$  has a left adjoint, called sheafification.

Eg The sheafification of  $U \mapsto \Lambda$  is  $\underline{\Lambda}$ .

Notation  $\text{Shv}(X) = \text{sheaves on } X \text{ valued in } \text{Ab}$ .

Fact  $\text{Shv}(X)$  is an abelian category with enough injectives

This means we can define cohomology groups  $H^n(X, F)$  for  $F \in \text{Shv}(X)$  as follows:

- Take a resolution of  $F$  by injectives, i.e.

$$0 \rightarrow F \rightarrow I_0 \rightarrow I_1 \rightarrow \dots \quad \text{exact in } \text{Shv}(X) \text{ with } I_n \text{ injective}$$

- Take global sections of  $I_\bullet$ , i.e.

$$I_0(X) \rightarrow I_1(X) \rightarrow I_2(X) \rightarrow \dots$$

- $H^n(I_\bullet(X)) =: H^n(X, F)$ , independent of  $I_\bullet$ .

Thm If  $X$  is locally path connected, then,  $\Lambda$  an ab group

$$H_{\text{sing}}^n(X, \Lambda) \cong H^n(X, \underline{\Lambda})$$

Idea Recall

$$H_{\text{sing}}^n(X, \Lambda) = H^n(C^\bullet(X, \Lambda))$$

with  $C^\bullet(X, \Lambda)$  the complex of cochains. For each  $n$ , let

$$C^n(X, \Lambda) = \text{sheafification of the functor } U \mapsto \{\Lambda\text{-valued cochains in } U\}$$

If  $X$  is locally path connected, then

$$\underline{\Lambda} \rightarrow C^0(X, \Lambda) \rightarrow C^1(X, \Lambda) \rightarrow \dots$$

is a resolution of  $\underline{\Lambda}$  by flabby sheaves (i.e.  $F$  is flabby if for every  $U \subseteq V$ ,  $\text{res}_U^V: F(V) \rightarrow F(U)$  is surjective)

$$\Rightarrow H^n(X, \underline{\Lambda}) = H^n(\Gamma(X, C^\bullet(X, \Lambda))) \\ = H^n(C^\bullet(X, \Lambda)) = H_{\text{sing}}^n(X, \Lambda).$$

Problem Still not good enough for alg geo, since if  $X$  is only irred top space, then

$$\underline{\Lambda}(U) = \Lambda \quad \text{for all open } U \subseteq X$$

$$\Rightarrow \underline{\Lambda} \text{ is flabby}$$

$$\Rightarrow H^n(X, \underline{\Lambda}) = 0 \quad \forall n > 0.$$

Solution Allow more general maps than inclusions  $U \subseteq X$ .

Def Let  $\mathcal{C}$  be a category. A Grothendieck (pre)topology on  $\mathcal{C}$  is the data of a distinguished family of maps  $(U_i \rightarrow U)_{i \in I}$  for every object  $U$  of  $\mathcal{C}$ , called coverings, satisfying the following axioms

1. For any covering  $(U_i \rightarrow U)_{i \in I}$  and morphism  $V \rightarrow U$  in  $\mathcal{C}$ , the fibre products  $U_i \times_U V$  exist and

$$(U_i \times_U V \rightarrow V)_{i \in I} \text{ is a covering}$$

2. If  $(U_i \rightarrow U)_{i \in I}$  is a covering and for each  $i \in I$ ,  $(V_{ij} \rightarrow U_i)_{j \in J_i}$  is a covering, then

$$(V_{ij} \rightarrow U)_{i,j} \text{ is a covering}$$

3.  $(U \xrightarrow{\text{id}} U)$  is a covering.

A category  $\mathcal{C}$  with a Groth topology is called a site.

Eg If  $X$  is a top space,  $\mathcal{C} = \text{Open}(X)$ , and covering = open cover. Here

$$U_i \times_U V = U_i \cap V.$$

Eg Eg  $X = \text{scheme}$ , this is called the Zariski site of  $X$ , denoted  $X_{\text{Zar}}$ .

Eg (The étale site) Let  $X$  be a scheme. The étale site of  $X$ , denoted  $X_{\text{ét}}$ , has objects étale maps  $U \rightarrow X$ , morphisms

$$U \rightarrow V \text{ s.t. } \begin{array}{ccc} U & \rightarrow & V \\ & \searrow & \swarrow \\ & X & \end{array} \text{ commutes}$$

and  $(U_i \xrightarrow{\varphi_i} U)_{i \in I}$  is a covering if  $U = \bigcup_{i \in I} \varphi_i(U_i)$ .

Def A morphism of schemes  $X \xrightarrow{f} Y$  is étale if

1.  $f$  is locally of finite presentation, i.e. for every  $x \in X$ , there is an open affine  $U = \text{Spec } A \subseteq X$  containing  $x$  such that  $f$  factors as

$$f|_U: \text{Spec } A \rightarrow \text{Spec } B \subseteq Y \text{ and}$$

$$A \cong B[x_1, \dots, x_n] / (g_1, \dots, g_m)$$

2.  $f$  is flat.

3. For all  $y \in Y$ ,  $X_y := X \times_Y \{y\}$  is  $\cong \coprod_{i \in I} \text{Spec } k_i$  with each  $k_i / k(y)$  a sep fld ext.

Ex Finite fld ext  $K/k$  is étale  $\Leftrightarrow$  it is separable

Thm Let  $X \rightarrow Y$  is a morph of quasiproj vars /  $\mathbb{C}$ . Then

$$X^{an} = X(\mathbb{C}) \xrightarrow{\text{with an top}} Y^{an} = Y(\mathbb{C})$$

is a local iso  $\Leftrightarrow X \rightarrow Y$  is étale.

Moral  $\sum$  wj étale  $\Leftrightarrow$  covering spaces.

Properties 1. Open immersions are étale.

2. Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$ . Then

(a)  $f, g$  étale  $\Rightarrow g \circ f$  is étale

(b)  $g \circ f, g$  étale  $\Rightarrow f$  is étale

3.  $f: X \rightarrow X'$  and  $g: Y \rightarrow Y'$  are étale morphisms of schemes /  $S$ ,

then  $f \times_S g: X \times_S Y \rightarrow X' \times_S Y'$  is étale

4.  $f: X \rightarrow Y$  is étale  $\Leftrightarrow$  for all  $x \in X$ ,  $\exists$  open affine nbhds  $x \in \text{Spec } A = U \subseteq X$  and  $\text{Spec } B = V \subseteq Y$  with  $f(U) \subseteq V$  and st.

$\exists$  presentation

$$A \cong B[x_1, \dots, x_n] / (f_1, \dots, f_n)$$

with  $\det\left(\frac{\partial f_i}{\partial x_j}\right)$  is a unit in the local ring at  $x$ .